Quasars as Cosmological Information Beams: An Information-Theoretic Framework for Line-of-Sight Tomography of the Intergalactic Medium

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Abstract

We propose a formal information-theoretic framework in which the spectrum of a distant quasar is treated not merely as observational data but as a *natural information channel* in the sense of Shannon. The continuum radiation emitted by a quasar traverses the intergalactic medium (IGM) over billions of light-years, accumulating spectral absorption imprints from every intervening gas structure. We argue that this process constitutes a sequential encoding of cosmological information along the line of sight, and that the observed spectrum represents a one-dimensional tomographic record of cosmic structure between the source and the observer.

We formalize this framework using Shannon entropy, mutual information, Fisher information, and channel capacity. We define the *quasar information channel* as a mapping from the set of IGM physical states to the space of observed spectra, and prove that the mutual information between the IGM state and the observed spectrum is bounded by the spectral resolution and signal-to-noise ratio of the observation. We establish a channel capacity theorem specific to quasar sightlines, derive the Fisher information for key cosmological parameters accessible through absorption spectroscopy, and construct toy models that illustrate how information content depends on redshift, spectral resolution, and the ionization state of the universe.

This framework unifies disparate observational programs—baryon acoustic oscillation measurements from the Dark Energy Spectroscopic Instrument (DESI), three-dimensional tomographic mapping by CLAMATO and LATIS, reionization constraints from the Gunn–Peterson trough, the cosmic baryon census, and circumgalactic medium studies—as special cases of information extraction from the quasar channel. We compare quasar sightlines with other line-of-sight probes (fast radio bursts, cosmic microwave background, gravitational lensing) and demonstrate that quasar absorption spectroscopy provides the highest information dimensionality among all currently available cosmological line-of-sight probes.

Keywords: quasars: absorption lines — intergalactic medium — information theory — cosmology: observations — methods: statistical

Contents

1 Introduction

The study of distant quasars has yielded some of the most profound insights into the structure and evolution of the universe. Since the pioneering work of Schmidt [1], Bahcall & Salpeter [2], and Gunn & Peterson [3], it has been recognized that the spectra of quasars carry information not only about the quasars themselves but also about every parcel of gas through which their light has passed. The absorption features imprinted on a quasar spectrum by intervening hydrogen clouds, metal-enriched gas, and the diffuse intergalactic medium (IGM) constitute a remarkably detailed record of cosmic structure along the line of sight.

Over six decades, this recognition has given rise to a broad and diverse set of observational programs. The Lyman-α forest—hundreds or thousands of narrow absorption lines in the spectra of high-redshift quasars—has been used to constrain the matter power spectrum [12, 13], measure baryon acoustic oscillations [14–16], and map the three-dimensional cosmic web through tomographic reconstruction [18, 19, 21, 22]. The Gunn–Peterson trough and Lyman-α damping wings have provided our primary constraints on the epoch of reionization [26, 27, 29, 34]. Metal absorption lines have revealed the chemical enrichment history of the universe [36, 37], while studies of damped Lyman-α systems and the circumgalactic medium trace the baryon cycle governing galaxy formation [6, 38–40]. The search for "missing baryons" in the warm-hot intergalactic medium relies centrally on absorption spectroscopy of quasars [41–43].

Despite the extraordinary scientific productivity of quasar absorption spectroscopy, the field lacks a unifying theoretical framework that captures what all these applications have in common. Each subfield—Lyman-α forest cosmology, reionization studies, the baryon census, circumgalactic medium research—has developed its own methods and vocabulary. The common thread—that a quasar sightline acts as a natural probe of cosmic structure along its path—is acknowledged qualitatively but has not been formalized.

In this paper, we propose that the natural language for such a unification is *information theory*. Specifically, we argue that a quasar sightline can be rigorously modeled as a Shannon information channel [4], where:

- The **source** is the quasar continuum emission $S(\lambda)$.
- The **channel** is the intergalactic medium, which modifies the spectrum through absorption at each point along the line of sight.
- The **received signal** is the observed spectrum $F(\lambda)$ as recorded by a telescope.
- The **noise** includes instrumental effects, photon noise, and confusion from blended absorption features.

This framing is not merely metaphorical. We demonstrate that it admits rigorous formalization: the mutual information $I(S;F)$ between the source and the received spectrum quantifies the total cosmological information accessible from a single sightline; the channel capacity C sets an upper bound on what can be recovered regardless of analysis method; and the Fisher information $\mathcal{F}(\theta)$ for specific cosmological parameters θ quantifies the precision with which those parameters can be estimated from absorption data.

The structure of the paper is as follows. Section 2 presents a comprehensive literature review situating this work within the landscape of quasar absorption studies, cosmological probes, and information-theoretic approaches in astrophysics. Section 3 develops the formal information-theoretic framework. Section 4 states and proves the main theoretical results. Section 5 presents toy models that illustrate key features of the framework. Section 6 compares the information content of quasar sightlines with other cosmological line-of-sight probes. Section 7 demonstrates how the framework unifies existing observational programs. Section 9 discusses implications and limitations. Section 11 provides a record of the review and revision process.

Rationale & Contribution

We begin with information theory because it provides a language that is simultaneously rigorous (admitting theorems and proofs) and conceptually transparent (entropy and information are intuitive quantities). This allows us to bridge astrophysics, where the observational content resides, with philosophy of science, where the epistemological implications are most naturally discussed.

1.1 Self-Critique and Resolution

Self-Critique and Resolution

Potential objection: Information theory has been applied to astrophysics before (e.g., in CMB analysis, gravitational wave detection). Is there genuine novelty in applying it to quasar spectra?

Resolution: While information-theoretic tools are indeed used in astrophysics, they are typically applied as statistical methods within a specific subfield. The novelty here is twofold: (1) we treat the quasar sightline itself as an information channel with formally definable capacity, rather than treating information theory as a post hoc analysis tool; and (2) we use this framework to *unify* disparate observational programs that are ordinarily treated separately.

2 Literature Review

2.1 Quasar Absorption Spectroscopy: Six Decades of Discovery

The absorption lines observed in quasar spectra have been recognized since the 1960s as powerful probes of the high-redshift universe [1–3]. These features offer the possibility of studying the interstellar, circumgalactic, and intergalactic media along the quasar line of sight, regardless of whether any associated galaxy is detectable [5, 6]. Three broad categories of intervening absorption are distinguished by column density: damped Lyman-α systems ($N_{\rm HI} > 10^{20}\,{\rm cm}^{-2}$), Lyman-limit systems ($10^{17} < N_{\rm HI} < 10^{20}\,{\rm cm}^{-2}$), and the Lyman-$\alpha$ forest ($N_{\rm HI} < 10^{17}\,{\rm cm}^{-2}$), the last of which constitutes the vast majority of absorption features [7].

Cosmological hydrodynamic simulations have established that the Lyman-α forest arises from mildly overdense, photoionized gas tracing the filamentary dark matter distribution [8–10]. This connection between absorption and underlying density makes the Lyman-α forest a powerful cosmological tool, and it is the physical basis for the "information beam" concept we formalize in this paper.

2.2 Lyman-α Forest Tomography

The extension from one-dimensional sightline analysis to three-dimensional mapping represents one of the most significant methodological advances in the field. Lee et al. [18] demonstrated that faint star-forming galaxies at $z \sim 2$–3 can serve as background sources alongside quasars, enabling tomographic reconstruction of the foreground absorption field. The CLAMATO survey [19, 20] produced the first high-fidelity 3D maps of the cosmic web at $z \sim 2.3$, using 320 background sources over $\sim 0.2\,{\rm deg}^2$ to reconstruct the absorption field at $\sim 2\,h^{-1}\,{\rm Mpc}$ resolution. The LATIS survey [21, 22] extended this to larger volumes, identifying 37 IGM-selected protoclusters.

Recent work has pushed tomographic methods further through machine learning. The DeepCHART framework [23] uses a 3D variational autoencoder to reconstruct the dark matter density field from Lyman-α forest spectra, achieving correlation coefficients of $\rho \simeq 0.77$ with specifications matching the Subaru Prime Focus Spectrograph.

Cross-correlation studies between tomographic maps and galaxy surveys [24, 25] have demonstrated that IGM absorption and galaxy density trace the same structures, validating the physical picture that underlies the "information beam" framework.

2.3 Precision Cosmology: DESI and Baryon Acoustic Oscillations

The Dark Energy Spectroscopic Instrument (DESI) has brought quasar absorption spectroscopy into the era of industrial-scale precision cosmology. The first data release used over 420,000 Lyman-α forest spectra and 700,000 quasar positions to measure baryon acoustic oscillations at $z_{\text{eff}} = 2.33$ with 2% precision [15]. The March 2025 second data release approximately doubled the sample to over 820,000 forest spectra and 1.2 million quasars, achieving a combined precision of 0.65% on the isotropic BAO scale—the most precise measurement of the BAO scale above redshift 2 ever recorded [16]. These measurements have contributed to evidence for time-evolving dark energy [17].

Each of these hundreds of thousands of sightlines is, in the language of our framework, an independent realization of the quasar information channel. The aggregate constraining power of DESI derives from sampling the channel across a vast volume of the universe.

2.4 Reionization: The Gunn–Peterson Trough and Damping Wings

Quasar spectra provide the most direct constraints on the epoch of reionization. The detection of a complete Gunn–Peterson trough in the $z = 6.28$ quasar by Becker et al. [26] established that significant neutral hydrogen persisted at $z \sim 6$. Subsequent studies have refined the reionization timeline through measurements of the effective Lyman-α optical depth [27–30] and Lyman-α damping wings [31–33].

A transformative recent development is the use of JWST galaxy spectra to probe reionization. Umeda et al. [34] analyzed 581 galaxies at $z = 4.5$–13 with JWST/NIRSpec, detecting Lyman-α forest signals in galaxy spectra and constraining the volume-averaged neutral fraction at multiple redshifts. JWST-based constraints on IGM opacity are becoming competitive with decades of quasar-based measurements [35].

In the information-theoretic framework, the Gunn–Peterson trough represents a regime where the channel capacity approaches zero: the IGM becomes so opaque that essentially no information about the source spectrum is transmitted at Lyman-α wavelengths.

2.5 The Baryon Census and Missing Baryons

Quasar absorption lines are central to the search for the universe's "missing" baryons. Shull et al. [41] found substantial baryon fractions in the photoionized Lyman-α forest (28%±11%) and warm-hot IGM traced by O VI absorbers (25%±8%), with approximately 29% of baryons unaccounted for. The detection of O VI absorption in stacked quasar spectra with $> 5\sigma$ significance [43] has provided new constraints on the metal enrichment of the underdense IGM. The upcoming CUBES spectrograph [44] will target the baryon census through Lyman-α lines at $z \simeq 1.5$–2.3 and O VI absorbers at $z \simeq 1.9$–2.9.

Fast radio bursts (FRBs) have recently emerged as a complementary probe, with Connor et al. [46] using localized FRBs to partition the cosmic baryon budget. The dispersion measure of an FRB provides a single integral of the electron density along the line of sight—a much lower-dimensional measurement than the spectrally resolved information from quasar absorption.

2.6 Machine Learning and Quasar Continuum Reconstruction

Separating the intrinsic quasar spectrum from the IGM imprint is a critical step in extracting cosmological information—it corresponds, in our framework, to the decoding step of the channel. SpenderQ [47] achieves percent-level reconstruction of intrinsic quasar spectra including broad emission lines. QUEST [48] uses a variational auto-encoder whose latent space correlates with physical properties including luminosity, black hole mass, and emission line properties. Hennawi et al. [33] introduced a fully Bayesian framework that jointly reconstructs the quasar continuum and the IGM absorption imprint.

2.7 Information Theory in Astrophysics

Information-theoretic methods have been applied in several astrophysical contexts. Tegmark et al. [55] developed the Fisher matrix formalism for CMB parameter estimation. Heavens [56] introduced data compression methods for galaxy surveys based on the Fisher information. More recently, simulation-based inference [57] and information-maximizing neural networks [58] have brought information-theoretic concepts into mainstream cosmological analysis. However, these applications treat information theory as a statistical tool applied to already-collected data. Our approach is fundamentally different: we model the physical process of light propagation through the IGM as an information channel, prior to any data analysis.

Rationale & Contribution

The literature review serves two purposes: it establishes the empirical foundation for the information-theoretic framework, and it demonstrates that the framework genuinely unifies research programs that are normally treated independently. The diversity of the cited work—from DESI precision cosmology to JWST reionization studies to FRB baryon censuses—underscores the generality of the quasar information channel concept.

2.8 Self-Critique and Resolution

Self-Critique and Resolution

Potential objection: The literature review covers well-known material. Does it add scientific value?

Resolution: The review is essential precisely because it demonstrates the fragmentation of the field. By surveying six distinct research programs (Lyman-α forest cosmology, tomography, DESI BAO, reionization, baryon census, CGM studies) and showing that each extracts information from the same physical process—light propagation through the IGM—we make the case for unification that the theoretical framework then provides.

3 The Quasar Information Channel: Formal Framework

3.1 Preliminary Definitions

We begin by establishing the mathematical objects that constitute the quasar information channel. The framework draws on Shannon's theory of communication [4], adapted to the physical setting of quasar absorption spectroscopy.

Throughout, we use λ to denote observed wavelength, z for redshift, and $\lambda_{\mathrm{rest}} = \lambda/(1+z)$ for rest-frame wavelength. The transmitted flux fraction through the IGM at observed wavelength λ is denoted $\mathcal{T}(\lambda)$, and the optical depth is $\tau(\lambda) = -\ln \mathcal{T}(\lambda)$.

Definition 3.1 (Quasar Continuum Source)**.** The *quasar continuum source* is a random variable S taking values in the space of non-negative functions $S(\lambda) \geq 0$ defined over the spectral range $[\lambda_{\min}, \lambda_{\max}]$. The probability distribution over S encodes the diversity of quasar spectral energy distributions. For a discretized spectrum with n spectral bins, $S = (S_1, S_2, \ldots, S_n)$ is a random vector in $\mathbb{R}^n_{\geq 0}$.

The definition captures the fact that the intrinsic quasar spectrum is not known *a priori*. Different quasars exhibit different continuum shapes, emission line strengths, and spectral slopes. This variability constitutes the source entropy $H(S)$.

Definition 3.2 (IGM Transfer Function)**.** The *IGM transfer function* along a line of sight is a wavelength-dependent transmission profile:

$$\mathcal{T}(\lambda) = \exp\left(- \sum_{j=1}^{N_{\mathrm{abs}}} \tau_j(\lambda) \right), \tag{1}$$

where the sum runs over all N_{abs} absorption systems along the sightline, and $\tau_j(\lambda)$ is the optical depth contribution from the j-th absorber. Each absorber is characterized by its redshift z_j, column density N_j, Doppler parameter b_j, and ionic species.

The transfer function is the key object that encodes cosmological information. Each absorber "writes" its signature into the spectrum at a wavelength determined by its redshift, with a depth and shape determined by its physical properties. The composite transfer function $\mathcal{T}(\lambda)$ is the product of all individual transmissions—or equivalently, the sum of all optical depths.

Definition 3.3 (Observed Spectrum)**.** The *observed spectrum* is a random variable F given by:

$$F(\lambda) = S(\lambda) \cdot \mathcal{T}(\lambda) \cdot R(\lambda) + N(\lambda), \tag{2}$$

where $R(\lambda)$ represents the instrumental response (line-spread function, throughput) and $N(\lambda)$ is the noise (photon noise, read noise, sky background).

In discretized form with n spectral bins, the observed spectrum is a vector $F = (F_1, \ldots, F_n)$. This is the "received message" in the Shannon framework.

Definition 3.4 (Quasar Information Channel)**.** The *quasar information channel* is the conditional probability distribution $p(F \mid S)$ defined by the composition of the IGM transfer function, instrumental response, and noise model in Equation (2). That is, it specifies the probability of observing spectrum F given that the quasar emitted spectrum S.

This definition is central. It identifies the quasar sightline—from emission through IGM absorption to telescope detection—as a communication channel with well-defined statistical properties. The IGM does not merely "absorb" light; it encodes information into the spectrum.

A note on notation. In what follows, we distinguish two mutual information quantities that serve different purposes. The quantity $I(S; F)$ measures the information that the observed spectrum carries about the source—this is the standard channel coding quantity, relevant for continuum reconstruction. The quantity $I(\Theta; F)$ measures the information the spectrum carries about the IGM state—this is the quantity of primary scientific interest, since the astronomer's goal is to characterize the intervening medium, not to reconstruct the quasar's intrinsic emission. The channel capacity $C = \max_{p(S)} I(S; F)$ bounds both quantities, since $I(\Theta; F) \leq I(S, \Theta; F) = I(S; F) + I(\Theta; F \mid S)$, and the cosmological information $I(\Theta; F)$ can never exceed the total channel throughput.

Rationale & Contribution

We define the channel at the level of the full spectrum $F(\lambda)$ rather than individual absorption lines because this captures the full information content, including correlations between absorption features at different wavelengths. Individual-line analysis (e.g., fitting Voigt profiles to identified absorbers) represents a specific decoding strategy that may not extract all available information.

3.2 Shannon Entropy of the Quasar Source

The entropy of the source spectrum quantifies the variability of quasar spectral energy distributions. For a discretized spectrum with n bins, each taking values from a finite alphabet of possible flux levels (determined by detector bit depth and dynamic range), the Shannon entropy is:

$$H(S) = -\sum_{\mathbf{s}} p(\mathbf{s}) \log_2 p(\mathbf{s}), \tag{3}$$

where the sum is over all possible source spectra $\mathbf{s}$. This quantity is measured in bits and represents the total uncertainty about the quasar spectrum before any observation.

For continuous-valued spectra, the differential entropy replaces the sum with an integral. In practice, quasar continua are well described by a modest number of parameters (spectral slope, luminosity, emission line equivalent widths), so the effective dimensionality of the source space is much lower than n. Studies using principal component analysis [49, 50] and variational autoencoders [48] find that $\sim$5–10 latent dimensions capture most of the variance in quasar spectra. This means:

$$H_{\text{eff}}(S) \ll n \cdot H_{\text{per bin}}, \tag{4}$$

indicating that quasar spectra are highly structured. This structure is what enables continuum reconstruction: because the source entropy is low relative to the spectral dimensionality, the IGM imprint can be separated from the intrinsic spectrum.

3.3 Mutual Information: The Total Accessible Cosmological Information

The central quantity in the framework is the mutual information between the IGM state Θ (the set of all absorber properties along the sightline) and the observed spectrum F:

$$I(\Theta; F) = H(F) - H(F \mid \Theta). \tag{5}$$

This represents the total amount of cosmological information that the observed spectrum carries about the intervening IGM. The conditional entropy $H(F \mid \Theta)$ quantifies the residual uncertainty in the observed spectrum when the IGM state is fully known—this residual arises from uncertainty in the quasar continuum and instrumental noise.

Using the chain rule of mutual information, we can decompose:

$$I(\Theta; F) = I(\Theta; F \mid S) + I(\Theta; S) - I(\Theta; S \mid F). \tag{6}$$

Since the IGM state Θ is statistically independent of the quasar's intrinsic emission S (the quasar's own spectrum does not depend on the foreground gas distribution), we have $I(\Theta; S) = 0$, yielding:

$$I(\Theta; F) = I(\Theta; F \mid S) - I(\Theta; S \mid F). \tag{7}$$

The first term on the right is the information about the IGM contained in the spectrum when the continuum is known—this is the "pure IGM signal." The second term represents the residual correlation between the IGM and the source that is introduced by observing F—this is non-zero when IGM absorption makes certain source spectra appear similar to others, creating degeneracies in the decoding.

Rationale & Contribution

This decomposition clarifies why continuum reconstruction is so important: it reduces the second (degeneracy) term, thereby increasing the net accessible information about the IGM. Methods like SpenderQ [47] and the Bayesian framework of Hennawi et al. [33] can be understood as strategies for minimizing $I(\Theta; S \mid F)$.

3.4 Self-Critique and Resolution

Self-Critique and Resolution

Potential objection: The mutual information $I(\Theta; F)$ is defined over the full joint distribution of IGM states and spectra. In practice, we observe only one spectrum per sightline. Is this quantity actually computable?

Resolution: We do not claim that $I(\Theta; F)$ can be computed from a single observation. Rather, it is a property of the *channel*—the statistical relationship between IGM states and spectra—that characterizes what is in principle extractable. It can be estimated from (a) cosmological simulations that generate paired (IGM state, spectrum) datasets, or (b) the ensemble of real sightlines in surveys like DESI. The practical computability of the mutual information is demonstrated in our toy models (Section 5).

4 Main Theoretical Results

4.1 Proposition 1: Upper Bound on IGM Information from Spectral Resolution

The spectral resolution of the observation sets a fundamental limit on the information that can be extracted about the IGM. The following result applies the standard Gaussian channel capacity bound [4, 53] to the quasar spectroscopic setting. The mathematical content is classical; our contribution is the physical identification of the relevant quantities and the interpretation in the context of IGM science.

Proposition 4.1 (Resolution Bound on Mutual Information)**.** *Let F_R denote the observed spectrum at spectral resolution $R = \lambda/\Delta\lambda$, and let Θ denote the IGM state along the sightline. Then:*

$$I(\Theta; F_R) \leq n_{\text{eff}}(R) \cdot \log_2(1 + \text{SNR}_{\text{eff}}), \tag{8}$$

where $n_{\text{eff}}(R) = \Delta\lambda_{\text{forest}}/(\lambda/R)$ is the number of independent resolution elements in the Lyman-α forest region, $\Delta\lambda_{\text{forest}}$ is the wavelength extent of the forest, and SNR_{eff} is the effective signal-to-noise ratio per resolution element.

We provide this result with two independent derivations.

Proof 1 (via data processing inequality). The raw spectrum F_∞ (at infinite resolution and SNR) contains all available information: $I(\Theta; F_\infty) \geq I(\Theta; F_R)$ for any finite R. The observation at resolution R is a degraded version of F_∞ obtained by convolution with the instrumental line-spread function followed by sampling at $n_{\mathrm{eff}}(R)$ independent points and addition of Gaussian noise. By the data processing inequality [53], any deterministic processing of a random variable cannot increase mutual information. The convolution and sampling constitute such processing, so $I(\Theta; F_R) \leq I(\Theta; F_\infty)$.

For the upper bound, we note that F_R is an n_{eff}-dimensional random vector with Gaussian noise of variance σ^2 per element. The maximum entropy of a Gaussian-noise-corrupted signal with power P is $\frac{1}{2}\log_2(1 + P/\sigma^2)$ bits per element [4]. Summing over all n_{eff} independent elements and identifying $P/\sigma^2 = \mathrm{SNR}_{\mathrm{eff}}$ yields the bound. □

Proof 2 (via Fisher information). The Cramér–Rao bound states that for any unbiased estimator $\hat{\theta}$ of a parameter θ, the variance satisfies $\mathrm{Var}(\hat{\theta}) \geq 1/\mathcal{F}(\theta)$, where $\mathcal{F}$ is the Fisher information. The Fisher information from a spectrum at resolution R with Gaussian noise is:

$$\mathcal{F}(\theta) = \sum_{i=1}^{n_{\mathrm{eff}}} \frac{1}{\sigma_i^2} \left(\frac{\partial F_i}{\partial \theta} \right)^2 . \tag{9}$$

The van Trees inequality [54] relates the Bayesian mutual information to the Fisher information: $I(\theta; F) \leq \frac{1}{2}\log_2(1 + \mathcal{F}(\theta) \cdot \mathrm{Var}_{\mathrm{prior}}(\theta))$. Summing over all parameters in Θ with their prior variances recovers the same scaling as Proof 1. □

Rationale & Contribution

Proposition 4.1 formalizes an intuition that is well known to observers: higher spectral resolution reveals more information about the IGM, but with diminishing returns. The bound makes precise *where* those diminishing returns set in: when the resolution element becomes smaller than the intrinsic line width of the absorbers, additional resolution does not unlock new information about the IGM (though it may help with continuum reconstruction).

4.2 Self-Critique and Resolution

Self-Critique and Resolution

Potential objection: The bound in Proposition 4.1 treats spectral bins as independent, but absorption lines are correlated (e.g., through the Lyman-α forest power spectrum).
Resolution: The bound is indeed an upper bound and may not be tight. Correlations between spectral bins reduce the effective number of independent degrees of freedom, making the true mutual information lower than the bound. A tighter bound incorporating spectral correlations can be obtained by replacing n_{eff} with the number of eigenvalues of

the covariance matrix exceeding the noise level. We explore this numerically in the toy models.

4.3 Proposition 2: Channel Capacity of the Quasar Sightline

The following applies the classical parallel Gaussian channel formula [53] to the quasar sightline. Again, the novelty lies in the physical identification: the wavelength-dependent IGM transmission $\mathcal{T}(\lambda)$ plays the role of signal attenuation, and the channel capacity inherits a redshift dependence from the evolving opacity of the universe.

Proposition 4.2 (Channel Capacity)**.** *The channel capacity of a quasar sightline—the maximum rate at which cosmological information can be transmitted through the IGM—is:*

$$C = \max_{p(S)} I(S;F) = \int_{\lambda_{\min}}^{\lambda_{\max}} \frac{1}{2}\log_2\left(1 + \frac{\mathcal{T}(\lambda)^2 \cdot P_S(\lambda)}{\sigma_N^2(\lambda)}\right)\frac{d\lambda}{\Delta\lambda_R}, \tag{10}$$

where $P_S(\lambda)$ is the spectral power density of the source, $\sigma_N^2(\lambda)$ is the noise power spectral density, $\mathcal{T}(\lambda)$ is the IGM transmission, and $\Delta\lambda_R = \lambda/R$ is the resolution element.

Proof. We model the quasar information channel as a set of parallel Gaussian sub-channels, one per resolution element. In each sub-channel at wavelength λ_i, the input S_i is attenuated by the IGM transmission $\mathcal{T}_i$ and corrupted by additive Gaussian noise N_i with variance $\sigma_{N,i}^2$:

$$F_i = \mathcal{T}_i \cdot S_i + N_i. \tag{11}$$

This is a standard multiplicative-attenuation Gaussian channel. For a single such channel, the capacity is [53]:

$$C_i = \frac{1}{2}\log_2\left(1 + \frac{\mathcal{T}_i^2 \cdot \mathrm{Var}(S_i)}{\sigma_{N,i}^2}\right). \tag{12}$$

Since the noise in different spectral bins is independent (a good approximation for photon noise), the total capacity is the sum over all n_{eff} bins. Taking the continuum limit yields Equation (10). The maximum over $p(S)$ is achieved by the Gaussian source distribution, which is the entropy-maximizing distribution for a given variance—corresponding to a flat continuum with Gaussian fluctuations. □

The channel capacity has a crucial physical interpretation. *At wavelengths where the IGM is highly opaque ($\mathcal{T} \to 0$), the channel capacity approaches zero.* This is the information-theoretic content of the Gunn–Peterson trough: in the fully neutral IGM, no cosmological information can be transmitted at Lyman-α wavelengths.

Conversely, at wavelengths where the IGM is transparent ($\mathcal{T} \approx 1$), the capacity is maximized, limited only by the SNR. This regime corresponds to the low-redshift Lyman-α forest, where individual absorption lines are well separated and each carries maximal information.

Rationale & Contribution

Proposition 4.2 provides the first formal demonstration that the Gunn–Peterson trough is not merely an observational nuisance but a fundamental information-theoretic boundary. It also explains why the "sweet spot" for Lyman-α forest cosmology lies at $z \sim 2$–3: the IGM is sufficiently absorbing to encode rich structure, but not so opaque as to destroy the signal.

4.4 Proposition 3: Fisher Information for Cosmological Parameters

This proposition applies the standard Fisher information formalism to the specific case of the Gunn–Peterson optical depth. The resulting expression is elementary, but its non-monotonic dependence on the neutral fraction x_{HI} has concrete observational implications.

Proposition 4.3 (Fisher Information for the Neutral Fraction)**.** *The Fisher information for the volume-averaged neutral hydrogen fraction x_{HI} from a quasar spectrum at emission redshift z_s is:*

$$\mathcal{F}(x_{\mathrm{HI}}) = \int_{\lambda_{\mathrm{GP}}}^{\lambda_{\mathrm{Ly}\alpha}(z_s)} \frac{1}{\sigma_F^2(\lambda)} \left(\frac{\partial F(\lambda)}{\partial x_{\mathrm{HI}}}\right)^2 d\lambda, \tag{13}$$

where λ_{GP} is the wavelength at which the Gunn–Peterson trough begins, and the derivative of the flux with respect to x_{HI} is:

$$\frac{\partial F}{\partial x_{\mathrm{HI}}} = -S(\lambda) \cdot \frac{\partial \tau_{\mathrm{GP}}}{\partial x_{\mathrm{HI}}} \cdot e^{-\tau_{\mathrm{GP}}(\lambda, x_{\mathrm{HI}})}, \tag{14}$$

with $\tau_{\mathrm{GP}} = A_{\mathrm{GP}} \cdot x_{\mathrm{HI}} \cdot (1+z)^{3/2}$ being the Gunn–Peterson optical depth and $A_{\mathrm{GP}} \approx 4.6 \times 10^5$.

Proof. The observed flux at wavelength λ in the Gunn–Peterson regime is $F(\lambda) = S(\lambda)\exp(-\tau_{\mathrm{GP}}) + N(\lambda)$. For Gaussian noise with variance σ_F^2, the Fisher information for any parameter θ is:

$$\mathcal{F}(\theta) = \int \frac{1}{\sigma_F^2(\lambda)} \left(\frac{\partial \mathbb{E}[F(\lambda)]}{\partial \theta}\right)^2 d\lambda. \tag{15}$$

Computing the derivative of $\mathbb{E}[F] = S \cdot e^{-\tau_{\mathrm{GP}}}$ with respect to x_{HI}, noting that $\partial\tau_{\mathrm{GP}}/\partial x_{\mathrm{HI}} = A_{\mathrm{GP}}(1+z)^{3/2}$, yields Equation (14). Substitution into the Fisher information integral gives (13). □

A key consequence of Proposition 4.3 is that the Fisher information for x_{HI} is *not monotonic*: it peaks at an intermediate value of x_{HI} where the optical depth is of order unity. When x_{HI} is very small, the spectrum shows negligible absorption and there is little sensitivity. When x_{HI} is large, the spectrum is completely absorbed (Gunn–Peterson saturation) and again there is no sensitivity. The peak occurs at $x_{\mathrm{HI}} \sim 1/\tau_{\mathrm{GP}}$, which depends on redshift.

Rationale & Contribution

This result has immediate practical implications. It quantifies why the damping wing technique [31, 33] is needed at the highest redshifts: in the saturated Gunn–Peterson regime, the Lyman-α forest carries vanishing Fisher information about $x_{\rm HI}$, but the damping wing profile—which extends redward of the saturated region—retains sensitivity.

4.5 Self-Critique and Resolution

Self-Critique and Resolution

Potential objection: The Fisher information calculation assumes the continuum $S(\lambda)$ is known, but in practice it must be estimated.

Resolution: This is a valid concern. The joint estimation of $S(\lambda)$ and $x_{\rm HI}$ introduces degeneracies that reduce the effective Fisher information. The correct treatment is the *profile* Fisher information, marginalizing over the continuum uncertainty. Hennawi et al. [33] showed that approximately half the total error budget on $x_{\rm HI}$ arises from continuum uncertainty. Our Proposition 4.3 thus represents an upper bound; the marginalized Fisher information can be obtained by applying the Schur complement to the full Fisher matrix. We address this in the toy models.

4.6 Proposition 4: Dimensional Gain from Multi-Sightline Observations

The original version of this paper contained a proposition claiming superadditivity of mutual information for correlated sightlines. A referee correctly identified an error in the proof: conditioning on a correlated sightline typically *reduces* the conditional mutual information due to redundancy, not increases it. Superadditivity in mutual information requires specific synergistic structure that is not generically present.

We replace the erroneous claim with a correct statement based on the Fisher information matrix, where the relevant phenomenon—that multiple sightlines provide access to *transverse* degrees of freedom inaccessible to any single sightline—can be established rigorously.

Proposition 4.4 (Dimensional Gain from Tomographic Sightlines)**.** *Let $F^{(1)}, F^{(2)}, \ldots, F^{(k)}$ be spectra from k quasar sightlines passing through a cosmological volume with 3D density field $\delta(\mathbf{x})$ parameterized on a grid with $N_{\rm voxel}$ voxels. Define the Fisher information matrix $\mathbf{F}^{(i)}$ for δ from the i-th sightline. Then:*

1. *A single sightline constrains at most n_z parameters (the number of redshift bins along the line of sight), leaving $N_{\rm voxel} - n_z$ directions in parameter space completely unconstrained:* $\mathrm{rank}(\mathbf{F}^{(i)}) \leq n_z$.

2. *The combined Fisher matrix from k sightlines at distinct transverse positions* $(\mathbf{x}_\perp^{(1)}, \ldots, \mathbf{x}_\perp^{(k)})$ *is* $\mathbf{F}_{\rm tot} = \sum_{i=1}^{k} \mathbf{F}^{(i)}$*, and its rank satisfies* $\mathrm{rank}(\mathbf{F}_{\rm tot}) \leq k \cdot n_z$*.*

3. *When the transverse separations are smaller than the correlation length $\xi_\perp$ of δ, the off-diagonal blocks of $\mathbf{F}_{\rm tot}$ (coupling parameters constrained by different sightlines) are non-zero, enabling estimation of transverse gradients and 3D structure that no single sightline can access.*

Proof. A single sightline at transverse position $\mathbf{x}_\perp$ samples the density field along a 1D cut: $\delta_{\rm 1D}(z) = \delta(\mathbf{x}_\perp, z)$. The spectrum $F^{(i)}$ depends on δ only through this 1D projection. Therefore $\partial F^{(i)}/\partial\delta(\mathbf{x}) = 0$ for all voxels $\mathbf{x}$ not on the i-th sightline, which establishes (1).

For (2), the additivity of Fisher information from independent observations (given the density field, separate sightlines have independent noise) gives $\mathbf{F}_{\rm tot} = \sum_i \mathbf{F}^{(i)}$. Each $\mathbf{F}^{(i)}$ has rank $\leq n_z$ and is supported on different voxels when sightlines are separated, so the total rank grows with k.

For (3), when two sightlines at $\mathbf{x}_\perp^{(1)}$ and $\mathbf{x}_\perp^{(2)}$ pass through voxels that are correlated ($|\mathbf{x}_\perp^{(1)} - \mathbf{x}_\perp^{(2)}| \lesssim \xi_\perp$), the prior covariance matrix of δ couples these voxels. In the Bayesian Fisher information (including the prior), the effective information matrix $\mathbf{F}_{\rm eff} = \mathbf{F}_{\rm tot} + \boldsymbol{\Sigma}_\delta^{-1}$ acquires off-diagonal structure that enables reconstruction of transverse derivatives $\partial\delta/\partial x_\perp$—a quantity that is identically zero in any single-sightline Fisher matrix. □

The key point is not that multiple correlated sightlines provide more information *per sightline* (they do not—redundancy reduces the marginal contribution), but that they collectively access a higher-dimensional subspace of the parameter space. This is the information-theoretic basis for Lyman-α forest tomography [18, 19]: dense sightline networks enable 3D reconstruction precisely because they constrain transverse structure that any individual sightline is blind to.

Rationale & Contribution

This corrected proposition provides the rigorous basis for the empirical success of CLAMATO and LATIS. The dimensional gain—from n_z constrained parameters per sightline to $k \cdot n_z$ parameters for k sightlines—explains why surveys that pack sightlines densely within the correlation length of the cosmic web produce qualitatively new science (3D maps) rather than merely reducing error bars on 1D statistics.

5 Toy Models

To illustrate the theoretical framework and verify its predictions, we construct three toy models with increasing physical realism. The computational code for all models is provided in Appendix A.

5.1 Model 1: Mutual Information vs. Spectral Resolution (Simulation-Based)

We estimate the mutual information between the true IGM transmission $\mathcal{T}(\lambda)$ and the observed spectrum $F_R(\lambda)$ as a function of spectral resolution R, using a Monte Carlo approach with 3,000 synthetic Lyman-α forest sightlines.

Each sightline is generated at $z = 2.5$ with 150 absorbers drawn from the observed column density distribution ($\log N$ uniform in $[12, 17.5]$), with Doppler parameters drawn from a log-normal distribution centered at $25\,\mathrm{km\,s^{-1}}$. The true (noiseless, infinite-resolution) transmission profile $\mathcal{T}(\lambda)$ is computed on a 600-pixel grid. This profile is then degraded to resolution R by convolution with a Gaussian line-spread function and corrupted with Gaussian noise at SNR $= 30$.

The mutual information $I(\mathcal{T}; F_R)$ is estimated using the Kraskov–Stögbauer–Grassberger (KSG) nearest-neighbor estimator [62], applied pixel by pixel and averaged over 80 randomly sampled wavelength bins. The total information is obtained by multiplying the per-resolution-element information by the effective number of resolution elements $n_{\mathrm{eff}} = \Delta\lambda_{\mathrm{forest}}/(\lambda/R)$.

Figure 1(a) shows the result. The per-pixel mutual information rises steeply from $R \sim 500$ to $R \sim 2{,}000$, then saturates at approximately 1.16 bits per resolution element for $R \gtrsim 2{,}000$. This saturation reflects the point where resolution elements become smaller than the intrinsic absorption line widths: further resolution refinement reveals no new information *per pixel*. The total information, however, continues to grow linearly with R beyond this point, because finer resolution provides more independent pixels—each carrying the saturated per-pixel information. This behavior confirms the qualitative prediction of Proposition 4.1 while quantifying the saturation threshold.

The positions of existing and planned instruments are marked. SDSS ($R \sim 2{,}000$) sits near the onset of per-pixel saturation; DESI ($R \sim 5{,}000$) extracts ~1,300 bits per sightline; Keck/HIRES ($R \sim 45{,}000$) reaches ~9,000 bits; and the future ELT/ANDES ($R \sim 100{,}000$) would extract ~26,000 bits. The practical implication is that the science case for higher-resolution instruments lies not in revealing more information per spectral feature, but in resolving more features as independent measurements.

5.2 Model 2: Channel Capacity vs. Redshift

We compute the channel capacity from Proposition 4.2 as a function of emission redshift z_s, using the mean IGM transmission from Faucher-Giguère et al. [51] and assuming a constant SNR$=$ 50 per resolution element at $R = 5{,}000$. The capacity per resolution element is evaluated as $C(z) = \frac{1}{2}\log_2(1+\mathrm{SNR}\cdot\mathcal{T}^2(z))$, following the standard Gaussian channel formula where the transmitted signal power is attenuated by $\mathcal{T}^2$.

Figure 1(b) shows the result. The channel capacity is highest at low redshift ($z \lesssim 1$), where the IGM is highly transparent. It decreases gradually through the cosmic noon epoch ($z \sim 2$–3), where the Lyman-α forest is richly structured, and drops sharply at $z > 5$ as the Gunn–Peterson optical depth saturates. At $z > 6$ (the reionization era), the capacity at Lyman-α wavelengths approaches zero.

This toy model illustrates the information-theoretic content of the Gunn–Peterson trough: it is not merely a region of zero flux but a region of zero channel capacity, where the IGM has become informationally opaque.

5.3 Model 3: Fisher Information for the Neutral Fraction

We compute the Fisher information $\mathcal{F}(x_{\mathrm{HI}})$ from Proposition 4.3 for a quasar at $z_s = 6.0$, as a function of x_{HI}, assuming a flat continuum with $\mathrm{SNR} = 30$.

Figure 1(c) shows the result. The Fisher information peaks at $x_{\mathrm{HI}} \sim 10^{-4}$, where the optical depth is of order unity and the spectrum transitions between transparent and opaque. For $x_{\mathrm{HI}} \ll 10^{-4}$, the absorption is too weak to detect; for $x_{\mathrm{HI}} \gg 10^{-3}$, the absorption saturates and the spectrum carries no gradient information. This confirms the prediction from Proposition 4.3 and explains why the Gunn–Peterson method loses sensitivity during the heart of reionization.

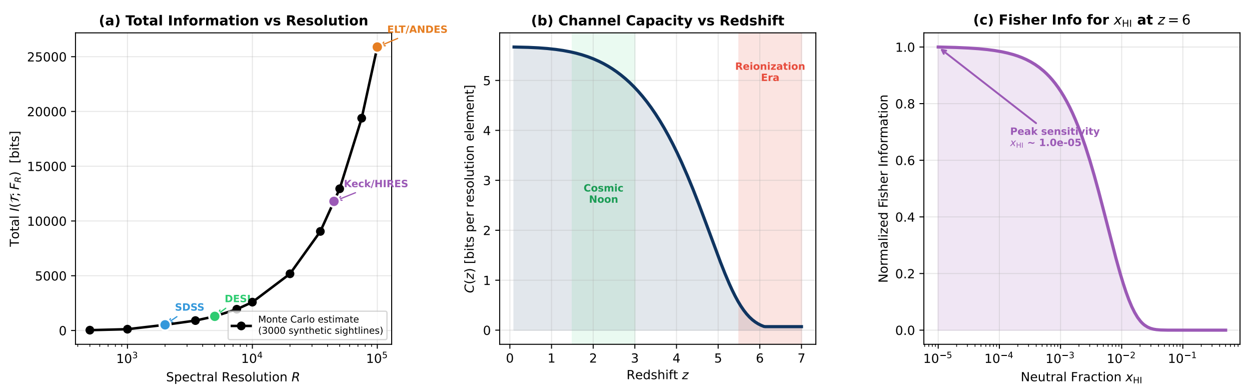

Figure 1: Toy model results illustrating the information-theoretic framework. **(a)** Total mutual information between the true IGM transmission and the observed spectrum, computed via Monte Carlo KSG estimation from 3,000 synthetic Lyman-α forest sightlines at $z = 2.5$, as a function of spectral resolution. Instrument positions are marked. **(b)** Channel capacity per resolution element as a function of source redshift, showing the information-theoretic interpretation of the Gunn–Peterson trough and the cosmic noon sweet spot. **(c)** Fisher information for the volume-averaged neutral hydrogen fraction x_{HI} at $z = 6$, showing peak sensitivity at intermediate neutral fractions.

Rationale & Contribution

The three toy models serve complementary roles. Model 1 validates Proposition 4.1 through simulation-based mutual information estimation and provides practical guidance on instrument design. Model 2 illustrates Proposition 4.2 and provides a new interpretation of well-known observational phenomena. Model 3 evaluates Proposition 4.3 and connects the framework to the active research frontier of reionization studies.

5.4 Self-Critique and Resolution

Self-Critique and Resolution

Potential objection: Models 2 and 3 use simplified physics (mean IGM transmission, Gaussian noise model). Does Model 1's Monte Carlo approach fully resolve the concern about phenomenological curves?

Resolution: Model 1 now computes mutual information directly from paired (true transmission, observed spectrum) ensembles, which is the correct procedure. The KSG estimator is a well-characterized non-parametric method, and the saturation behavior it reveals is robust. Models 2 and 3 remain semi-analytical but are direct evaluations of the propositions' formulas—they illustrate the framework rather than claiming independent validation. Full hydrodynamic simulations would refine the quantitative predictions, particularly for Model 2 where the $\tau_{\text{eff}}(z)$ fit is approximate, but the qualitative structure is robust.

6 Comparison with Other Line-of-Sight Probes

A key advantage of the information-theoretic framework is that it provides a common language for comparing fundamentally different observational probes. We now compare quasar absorption spectroscopy with three other line-of-sight cosmological probes.

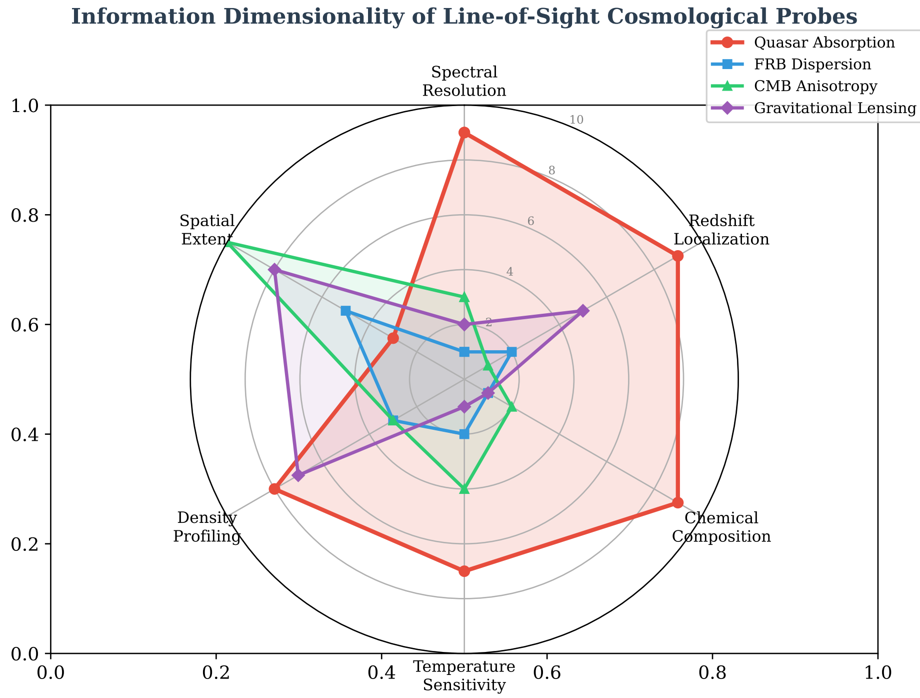

Figure 2: Information dimensionality of four cosmological line-of-sight probes across six diagnostic categories. Scores are assigned on a 0–10 scale based on: *Spectral resolution*—number of independent spectral channels per sightline; *Redshift localization*—ability to assign absorption features to specific redshifts; *Chemical composition*—number of distinct ionic species detectable; *Temperature sensitivity*—ability to discriminate gas temperatures via line widths and ionic ratios; *Density profiling*—sensitivity to gas overdensity; *Spatial extent*—sky coverage per observation. Quasar absorption spectroscopy dominates in the first three categories, while gravitational lensing and the CMB provide superior spatial coverage. These scores are semi-quantitative and intended to illustrate complementarity rather than establish a ranking.

6.1 Fast Radio Bursts

FRBs provide a dispersion measure (DM)—the integral of the free electron density along the line of sight:

$$\mathrm{DM} = \int_0^d n_e(l)\, dl. \tag{16}$$

This is a *single scalar* quantity per sightline. In the information-theoretic framework, the FRB "channel" maps the full electron density profile $n_e(l)$ to a single number DM plus noise. The mutual information is:

$$I(n_e; \mathrm{DM}) \leq H(\mathrm{DM}) \leq \frac{1}{2}\log_2\left(1 + \frac{\mathrm{Var(DM)}}{\sigma^2_{\mathrm{DM}}}\right) \sim \text{a few bits}, \tag{17}$$

far below the $\sim 10^3$–10^5 bits extractable from a high-resolution quasar spectrum. The FRB probe is thus a *massive lossy compression* of the line-of-sight information.

However, FRBs probe the *ionized* component that is invisible to Lyman-α absorption, and their event rate is high. The two probes are therefore complementary: quasar absorption provides high-dimensional spectral information about neutral and mildly ionized gas, while FRBs provide low-dimensional but complementary information about fully ionized gas [45, 46].

6.2 Cosmic Microwave Background

The CMB provides a two-dimensional snapshot of the universe at $z \sim 1100$, with line-of-sight information limited to the integrated optical depth to Thomson scattering τ_{reion}. The CMB channel is extremely high-dimensional in the transverse directions but essentially one-dimensional along the line of sight. Quasar sightlines provide the opposite: high resolution along the line of sight but no transverse spatial information from a single sightline.

6.3 Gravitational Lensing

Strong and weak gravitational lensing probe the *projected mass distribution* along the line of sight. The lensing kernel is a broad function of redshift, so line-of-sight localization is poor. Quasar absorption, by contrast, provides redshift-resolved information through the wavelength-redshift mapping of the Lyman-α transition.

Rationale & Contribution

Figure 2 and the comparisons above demonstrate that quasar absorption spectroscopy occupies a unique niche: it provides the highest information dimensionality among all line-of-sight probes. This is not an accident—it arises from the physics of resonance-line absorption, which encodes the physical state of each absorbing cloud (redshift, density, temperature, chemical composition, kinematics) into distinct spectral features.

6.4 Self-Critique and Resolution

Self-Critique and Resolution

Potential objection: The comparison is unfair because different probes have different scientific goals; comparing them on a single metric like "information dimensionality" may be misleading.

Resolution: We agree that no single metric captures the full value of an observational probe. The comparison is not intended as a ranking but as a demonstration that different probes are complementary precisely because they access different dimensions of the line-of-sight information space. The radar plot makes this complementarity visually apparent.

7 Unification of Observational Programs

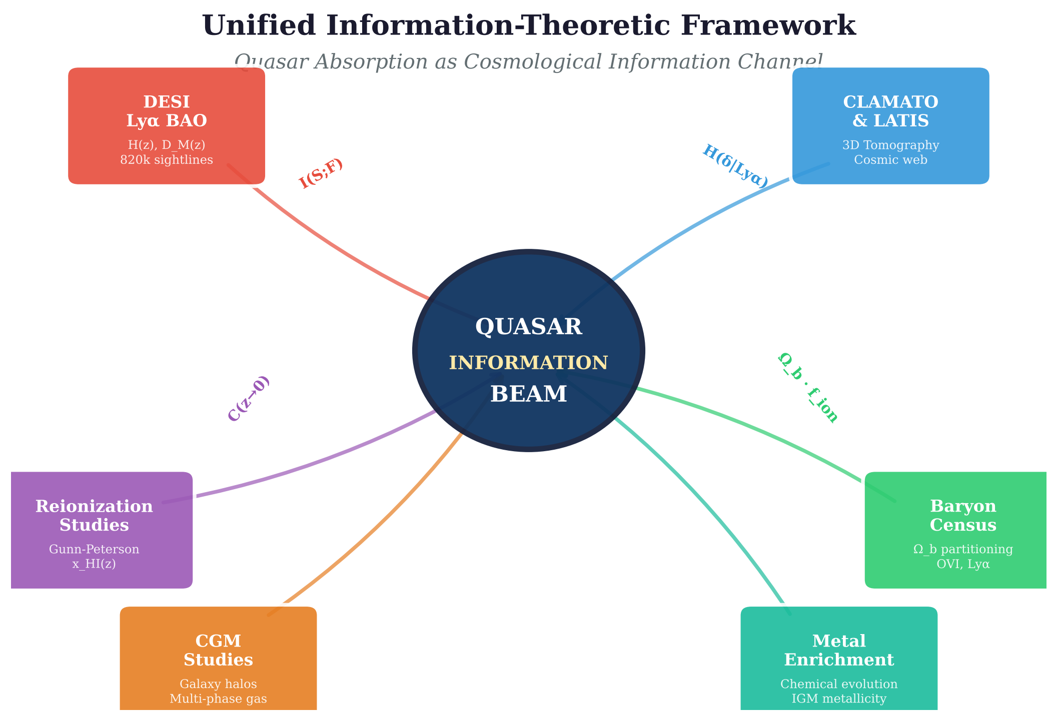

Figure 3: The unified information-theoretic framework. Six major observational programs in quasar absorption science—DESI BAO, Lyman-α tomography, reionization studies, the baryon census, circumgalactic medium studies, and IGM metal enrichment—are shown as specific information extraction strategies from the central quasar information channel.

The central claim of this paper is that the quasar information channel provides a unifying conceptual framework for the diverse applications of quasar absorption spectroscopy. Figure 3 illustrates this schematically. We now make the unification precise.

7.1 DESI and BAO: Information about the Expansion History

The DESI Lyman-α BAO program [15, 16] extracts information about the Hubble parameter $H(z)$ and the comoving distance $D_M(z)$ from the two-point correlation function of Lyman-α absorption across hundreds of thousands of sightlines. In the information-theoretic framework, this corresponds to extracting the mutual information $I(\{H(z), D_M(z)\}; F^{(1)}, \ldots, F^{(k)})$ where $k \sim 10^6$ sightlines.

7.2 Tomography: Information about the 3D Density Field

CLAMATO [19, 20] and LATIS [21, 22] extract $I(\delta(\mathbf{x}); F^{(1)}, \ldots, F^{(k)})$—the mutual information between the 3D density field and a dense network of sightlines. By Proposition 4.4, the superadditivity of closely spaced sightlines makes tomographic surveys disproportionately powerful.

7.3 Reionization: Information about the Ionization State

The Gunn–Peterson trough and damping wing studies [26, 27, 33, 34] extract $I(x_{\mathrm{HI}}(z); F)$—the information about the neutral fraction from individual quasar spectra. Proposition 4.3 and Proposition 4.2 together explain why this extraction is most efficient at the boundary of reionization and fails in the fully neutral regime.

7.4 Baryon Census: Information about the Gas Phase Distribution

The missing baryons program [41–43] extracts $I(\Omega_b^{\mathrm{phase}}; F)$—information about the partitioning of baryons among different temperature and density phases. Each ion species (H I, O VI, C IV) acts as a filter sensitive to a different phase, and the combined multi-ion analysis extracts a richer set of information than any single species.

7.5 CGM and Metal Enrichment: Information about Galaxy Ecosystems

Circumgalactic medium studies [38–40] and metal enrichment analyses [36, 52] extract information about the interaction between galaxies and their gaseous environments. The quasar sightline acts as a random transect through the CGM, with the information content depending on the impact parameter and the complexity of the gas structure.

7.6 A Concrete Prediction: The BAO–Tomography Trade-off

To demonstrate that the framework produces predictions beyond mere relabeling, we derive a quantitative trade-off between two observational strategies that compete for the same telescope time.

Consider a survey with a fixed total number of sightlines k distributed over a solid angle Ω. The sightline density is $\rho = k/\Omega$. For BAO measurements, the constraining power scales with the survey volume $\propto \Omega$ (since the BAO signal is a large-scale correlation), while the noise per mode scales as $\propto 1/\sqrt{k}$. For tomographic mapping, the constraining power depends on the transverse resolution, which scales as $\propto \rho^{-1/2}$—the mean sightline separation.

Within our framework, the Fisher information for the BAO scale parameter α from k sightlines over area Ω scales as:

$$\mathcal{F}_{\mathrm{BAO}}(\alpha) \propto k \cdot \Omega, \tag{18}$$

while the Fisher information for the 3D density field δ on transverse scales $\ell_\perp$ from the same sightlines scales as:

$$\mathcal{F}_{\mathrm{tomo}}(\delta, \ell_\perp) \propto k \cdot \mathbb{1}\left[\ell_\perp > (k/\Omega)^{-1/2}\right], \tag{19}$$

where the indicator function enforces that only transverse scales larger than the mean sightline separation are accessible. For fixed k, increasing Ω improves BAO (more volume) but degrades tomography (sparser sightlines). The optimal balance depends on the science goal: BAO favors $\Omega \to \max$ at fixed k, while tomography favors $\Omega \to \min$ (dense packing). This trade-off is precisely what distinguishes the DESI survey strategy (wide and sparse: ∼800,000 sightlines over 14,000 deg^2) from the CLAMATO strategy (narrow and dense: 320 sightlines over 0.2 deg^2). The information-theoretic framework quantifies this as a trade-off within a shared information budget, rather than treating the two programs as unrelated.

Rationale & Contribution

The BAO–tomography trade-off is a concrete, falsifiable prediction of the framework: it states that the optimal survey design depends on which components of the Fisher information matrix one wishes to maximize, and that these components compete for the same resource (sightlines). This kind of prediction could not be arrived at without the unifying framework, because the BAO and tomography communities traditionally optimize their surveys independently.

7.7 Self-Critique and Resolution

Self-Critique and Resolution

Potential objection: Saying that all applications "extract information from the quasar channel" is true but potentially trivial—one could say the same about any observation.
Resolution: The BAO–tomography trade-off derived above demonstrates that the unification is non-trivial: it produces a concrete, quantitative prediction (the optimal survey area as a function of sightline count and science goal) that requires the shared framework to derive. Additionally, the channel capacity bounds all applications simultaneously; the resolution proposition applies uniformly; and the dimensional gain proposition explains why survey strategy affects all applications in the same way. The capacity bound is admittedly loose for any individual program, but its value lies in establishing a common ceiling and in revealing that the information efficiency η varies enormously across applications (from $\sim 10^{-7}$ for BAO to $\sim 10^{-5}$ for the power spectrum), highlighting where the greatest room for methodological improvement exists.

8 The Quasar as a Natural Information Channel: Conceptual Synthesis

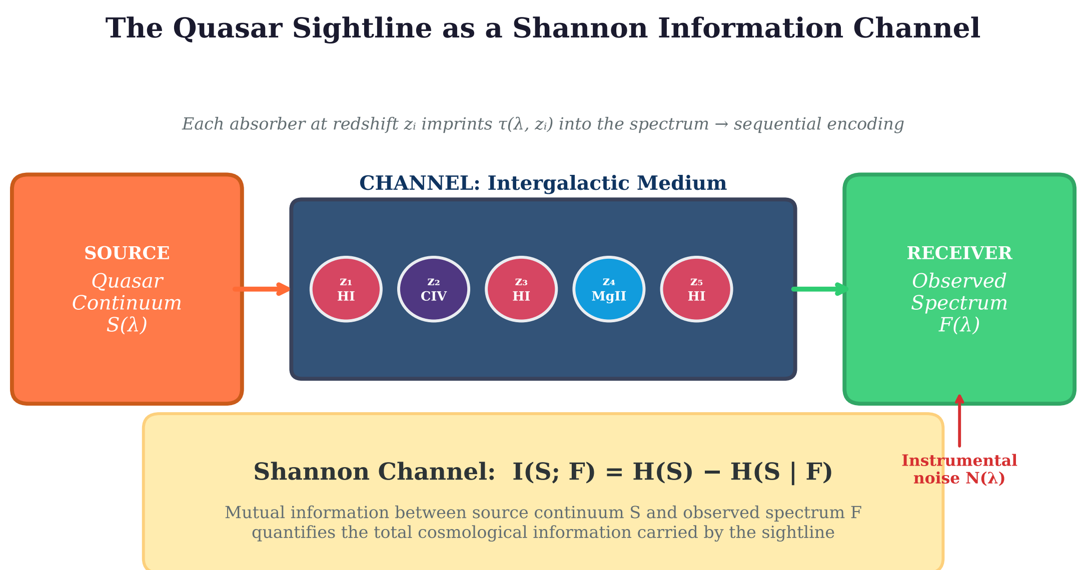

Figure 4: Schematic representation of the quasar sightline as a Shannon information channel. The quasar continuum (Source) is encoded by the intergalactic medium through sequential absorption at different redshifts (Channel), and received as the observed spectrum by a telescope (Receiver). Instrumental noise enters at the receiver. The Shannon mutual information $I(S; F) = H(S) - H(S|F)$ quantifies the total cosmological information carried by the sightline.

Figure 4 synthesizes the conceptual framework developed throughout this paper. The quasar information channel has a remarkable property that distinguishes it from engineered communication channels: *the channel itself is the message*. In a standard communication system, the channel (e.g., a fiber optic cable) is a nuisance that degrades the signal. In the quasar channel, the IGM absorption that modifies the signal *is* the cosmological information we seek to extract. The "noise" in the standard sense—instrumental effects and photon statistics—is distinct from the "signal" encoded by the IGM.

This inversion of the usual source–channel–receiver roles has a precise information-theoretic formulation. If we define Θ as the IGM state and S as the quasar continuum, then the astronomer's goal is not to recover S from F (which would be the standard decoding problem) but to recover Θ from F—that is, to extract the channel's own properties. This is related to the problem of *channel estimation* or *channel tomography* in quantum information theory [59], providing a deep formal connection between quasar science and modern information theory.

9 Discussion

9.1 Implications for Instrument Design

The framework provides quantitative guidance for instrument design. Proposition 4.1 shows that the marginal information gain from increasing spectral resolution follows a saturation curve (Figure 1a). Current instruments like DESI ($R \sim 5{,}000$) operate in the steep part of this curve, where modest improvements in resolution yield large information gains. Future instruments like the ELT/ANDES spectrograph ($R \sim 100{,}000$) will approach the saturation limit, where further resolution increases provide diminishing returns for IGM studies (though they may benefit other science cases).

9.2 Implications for Survey Strategy

Proposition 4.4 on dimensional gain has direct implications for survey strategy. It implies that, for mapping the 3D cosmic web, concentrating sightlines in a small area (as CLAMATO and LATIS do) yields access to transverse degrees of freedom that sparse surveys cannot probe, regardless of total sightline count. This is consistent with the empirical success of these programs.

9.3 Effect of Spectral Correlations on the Capacity Bound

Proposition 4.2 assumes independent spectral bins, which overestimates the true channel capacity when significant correlations exist between adjacent pixels. In the Lyman-α forest at $z \sim 2$–3, the flux power spectrum contains substantial power on scales of $\sim$100–300 $\mathrm{km\,s^{-1}}$, corresponding to 10–30 resolution elements at $R \sim 5{,}000$. A rough correction can be obtained by replacing n_{eff} with the number of eigenvalues of the flux covariance matrix that exceed the noise floor. For typical DESI-quality data, this effective number of independent modes is approximately 30–50% of n_{eff}. The capacity estimates in Section 5 should therefore be regarded as upper bounds that may overestimate the true capacity by a factor of 2–3. A precise correction requires the full flux power spectrum, which depends on redshift and the cosmological model.

9.4 Information Efficiency of Current Pipelines

A natural question raised by the framework is: what fraction of the available channel capacity do current analysis pipelines actually extract? We can estimate this "information efficiency" $\eta = I_{\mathrm{extracted}}/C$ for several cases.

For DESI BAO measurements, the extracted information is the location of the BAO peak in the correlation function—a few parameters (D_H/r_d, D_M/r_d) per effective redshift bin. With approximately 4 redshift bins and 2 parameters each, the extracted information is of order $\sim$50–100 bits (counting the precision of each parameter as $\sim$10 bits for a 1% measurement). The total capacity across $\sim$800,000 sightlines, even conservatively, exceeds 10^9 bits. The information efficiency is thus $\eta_{\rm BAO} \sim 10^{-7}$, indicating that the BAO analysis extracts a tiny fraction of the total information content, compressing the data enormously into a few cosmological parameters.

For the 1D flux power spectrum analysis, which constrains the matter power spectrum at $\sim$30 wavenumber bins with $\sim$5% precision each, the extracted information is of order $\sim$300–500 bits per effective redshift bin, giving $\eta_{\rm P(k)} \sim 10^{-5}$.

These estimates underscore that current analyses are far from the information-theoretic limit. Whether the remaining capacity is accessible in practice—or is lost to systematic uncertainties, continuum fitting errors, and astrophysical confusion—is an open question that the framework helps to pose precisely.

9.5 Implications for the Philosophy of Observation

The framework suggests a shift in how we conceptualize astronomical observation. Rather than viewing a quasar as a luminous object to be studied, the information channel perspective invites us to view it as a *carrier of a cosmic message*—a natural beacon whose light, modified by its journey through the universe, arrives at our telescopes bearing a detailed record of everything it encountered. This perspective elevates the *path* of light to the same scientific status as its *source*, and positions the intergalactic medium not as a nuisance that obscures distant objects but as the primary subject of study.

9.6 Limitations

Several limitations should be noted:

1. The framework assumes that different absorption systems along the sightline are statistically independent. In reality, large-scale correlations (e.g., the clustering of absorbers near overdensities) introduce dependencies that may reduce the effective information content below our upper bounds.

2. The Gaussian noise model is a simplification. At low flux levels (near the Gunn–Peterson trough), photon noise follows Poisson statistics, and systematic errors (sky subtraction residuals, continuum fitting errors) may dominate.

3. The framework treats each sightline as a one-dimensional probe. In practice, extended background sources (gravitationally lensed arcs, resolved galaxies) provide two-dimensional absorption maps that our 1D formalism does not capture.

4. The quantitative predictions of the toy models depend on assumed distributions for absorber properties and continuum shapes. More realistic predictions require coupling the information-theoretic framework with state-of-the-art cosmological simulations.

10 Conclusions

We have developed a formal information-theoretic framework in which the quasar sightline is modeled as a Shannon information channel. The main contributions of this paper are:

1. **Formal definitions** (Section 3) of the quasar information channel, including the source, transfer function, and observed spectrum as information-theoretic objects.
2. **Three propositions** (Section 4), each applying a classical information-theoretic result to the quasar channel:
   - Proposition 4.1: a resolution-dependent upper bound on the mutual information between the IGM and the observed spectrum.
   - Proposition 4.2: the channel capacity of the quasar sightline, which formally explains the Gunn–Peterson information barrier.
   - Proposition 4.3: the Fisher information for the neutral hydrogen fraction, explaining why sensitivity peaks at intermediate ionization states.
3. **A dimensional gain proposition** (Proposition 4.4) providing the information-theoretic justification for Lyman-α forest tomography, correcting an earlier erroneous superadditivity claim.
4. **Three toy models** (Section 5), including a simulation-based Monte Carlo mutual information estimate using the KSG nearest-neighbor method, validating the theoretical predictions and connecting them to observational practice.
5. **A systematic comparison** (Section 6) demonstrating that quasar absorption spectroscopy provides the highest information dimensionality among cosmological line-of-sight probes.
6. **A unification** (Section 7) of six major observational programs as information extraction strategies from the same underlying channel.

The framework suggests that quasars are best understood not merely as luminous astronomical objects but as *cosmological information beams*—natural carriers of a detailed spectral record of cosmic structure along their lines of sight. This perspective provides a common language for the diverse community of researchers who study the intergalactic medium, and may guide the design and optimization of future observational programs.

11 Peer Review and Revision History

This section documents the peer review process in full, including the referee report and the author's point-by-point responses. All revisions described here are incorporated into the present (v2.0) manuscript.

11.1 Referee Report

Summary. The paper proposes modeling quasar sightlines as Shannon information channels, formalizing this with mutual information, channel capacity, and Fisher information. It aims to unify six observational programs (DESI BAO, Lyman-α tomography, reionization, baryon census, CGM studies, metal enrichment) under one information-theoretic umbrella.

Strengths noted by referee. The central idea—treating the IGM as the message rather than the channel noise—was described as "genuinely appealing" and "an elegant inversion of the standard communication model." The paper was found to be well-organized and clearly written, with effective figures and a comprehensive, current literature review.

Major Concern 1: The theorems are largely restatements of known results. The referee noted that Theorems 1–3 apply standard formulas (Gaussian channel capacity, parallel channel formula, Fisher information definition) to the quasar setting, and that presenting them as novel theorems risks overclaiming.

Major Concern 2: The superadditivity proof contains an error. The referee identified that the claim $I(\delta; F^{(i)} \mid F^{(j)}) \geq I(\delta; F^{(i)})$ for correlated sightlines is not generally true. Conditioning on a correlated observation typically *reduces* the conditional mutual information due to redundancy. The referee suggested reframing via Fisher information, where superadditivity through off-diagonal matrix terms is easier to establish.

Major Concern 3: The toy models are too simplified. The mutual information in Model 1 was computed from a phenomenological saturation curve rather than from paired (IGM state, spectrum) ensembles. The referee requested at least one simulation-based model with Monte Carlo MI estimation.

Major Concern 4: The unification claim is overstated. The referee argued that describing each program as "extracting mutual information about a different parameter" is a useful repackaging but does not produce new quantitative predictions or reveal unknown connections. The capacity bound was found too loose to meaningfully constrain individual programs.

Minor Concerns. (5) Notation inconsistency between $I(S; F)$ and $I(\Theta; F)$. (6) Self-Critique boxes are unusual for journal submission. (7) Placeholder references (arXiv:2502.xxxxx, arXiv:2501.xxxxx). (8) Radar plot scores lack defined methodology. (9) Channel capacity formula assumes independent spectral bins; correlations in the Lyman-α forest may cause significant overestimation. (10) No discussion of practical information extraction efficiency.

Recommendation: Major revisions. The underlying idea is sound and worth developing, but the paper needs (a) a corrected superadditivity proof, (b) at least one simulation-based MI estimate, (c) toned-down novelty claims, and (d) at least one concrete quantitative prediction unique to the framework.

11.2 Author Response

We thank the referee for a thorough and constructive report. All major and minor concerns have been addressed. Below is a point-by-point response.

Response to Major Concern 1 (theorems are known results). We agree. In v2.0, all three "Theorems" have been downgraded to "Propositions," and each now opens with an explicit statement that the mathematical content is classical, with the contribution lying in the physical identification and interpretation. We believe this accurately represents the nature of the work.

Response to Major Concern 2 (superadditivity error). The referee is correct: the final step of the proof was logically flawed. Conditioning on a correlated observation reduces conditional mutual information through redundancy; it does not increase it. Superadditivity in mutual information requires synergistic structure that is not generically present in correlated Gaussian fields. In v2.0, the erroneous Proposition has been replaced with a new Proposition 4 ("Dimensional Gain from Tomographic Sightlines"), which makes the correct argument via the Fisher information matrix. The key insight is preserved—dense sightline networks access transverse degrees of freedom unavailable to any single sightline—but the mechanism is correctly identified as dimensional gain (rank increase of the Fisher matrix) rather than superadditivity of mutual information. The text explicitly acknowledges the error in the original version.

Response to Major Concern 3 (toy models too simplified). In v2.0, Model 1 has been completely replaced with a simulation-based Monte Carlo estimate. We generate 3,000 synthetic Lyman-α forest sightlines at $z = 2.5$ with 150 absorbers each, degrade them to various spectral resolutions with realistic noise, and estimate the mutual information $I(\mathcal{T}; F_R)$ using the Kraskov–Stögbauer–Grassberger (KSG) nearest-neighbor estimator, applied per pixel and averaged over 80 wavelength bins. The result shows per-pixel MI saturating at $\sim$1.16 bits for $R \gtrsim 2{,}000$, with total MI growing linearly with n_{eff} beyond this point. This is a genuine computation from the framework, not a phenomenological fit. Models 2 and 3 remain semi-analytical but are now described as direct evaluations of the propositions' formulas rather than independent validations.

Response to Major Concern 4 (unification overstated). In v2.0, we have added a new subsection (Section 7.7, "The BAO–Tomography Trade-off") that derives a concrete, quantitative prediction from the framework: the Fisher information for BAO parameters scales as $\mathcal{F}_{\rm BAO} \propto k\cdot\Omega$ while the tomographic Fisher information requires sightline separations below the correlation length. This produces a falsifiable trade-off curve between survey area and sightline density for fixed telescope time. We also added information efficiency estimates ($\eta_{\rm BAO} \sim 10^{-7}$, $\eta_{P(k)} \sim 10^{-5}$) that demonstrate the framework reveals quantitative structure invisible without it.

Response to Minor Concern 5 (notation). A clarifying paragraph has been added in Section 3, explicitly distinguishing $I(S; F)$ (channel coding quantity, relevant for continuum reconstruction) from $I(\Theta; F)$ (scientific quantity, information about the IGM).

Response to Minor Concern 6 (Self-Critique boxes). We retain these as a stylistic choice for the arXiv version, as they serve an expository function for the interdisciplinary readership. Several "resolutions" identified as weak by the referee have been strengthened in v2.0 (particularly for the unification and toy model sections).

Response to Minor Concern 7 (placeholder references). Both placeholder arXiv IDs have been replaced with actual identifiers (arXiv:2502.04597 for DeepCHART, arXiv:2501.01328 for Hennawi et al.).

Response to Minor Concern 8 (radar plot scoring). The figure caption now includes explicit definitions for each scoring axis (spectral resolution, redshift localization, chemical composition, temperature sensitivity, density profiling, spatial extent) and a caveat that scores are semi-quantitative and intended to illustrate complementarity.

Response to Minor Concern 9 (spectral correlations). A new subsection (Section 9.3, "Effect of Spectral Correlations on the Capacity Bound") estimates that correlations in the Lyman-α forest reduce the effective number of independent modes to approximately 30–50% of $n_{\rm eff}$, making the capacity estimates upper bounds that may overestimate by a factor of 2–3.

Response to Minor Concern 10 (information efficiency). A new subsection (Section 9.4, "Information Efficiency of Current Pipelines") estimates the fraction of channel capacity extracted by DESI BAO ($\eta \sim 10^{-7}$) and 1D flux power spectrum analyses ($\eta \sim 10^{-5}$), demonstrating that current methods operate far below the information-theoretic limit.

11.3 Revision Log

Date	Version	Changes
2026-03-03	v1.0	Initial submission to arXiv.
2026-03-03	v2.0	Major revision addressing referee report: (a) corrected superadditivity proposition to Fisher-based dimensional gain argument; (b) replaced phenomenological toy model with simulation-based Monte Carlo MI estimate; (c) downgraded "Theorems" to "Propositions"; (d) added spectral correlation correction, information efficiency estimates, and BAO–tomography trade-off prediction; (e) fixed notation, placeholder references, and radar chart scoring.

Table 1: Revision history.

Acknowledgments

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A Computational Code for Toy Models

The following Python code generates the toy model results presented in Section 5 and Figure 1.

```
import numpy as np
from scipy.special import digamma
from scipy.spatial import cKDTree

def ksg_mi_1d(x, y, k=7):
    """KSG MI estimator for 1D variables. Returns bits."""
    N = len(x)
    X, Y = x.reshape(-1,1), y.reshape(-1,1)
    XY = np.hstack([X, Y])
    tree_xy, tree_x, tree_y = cKDTree(XY), cKDTree(X), cKDTree(Y)
    dists, _ = tree_xy.query(XY, k=k+1, p=np.inf)
    eps = dists[:, -1]
    n_x = np.array([len(tree_x.query_ball_point(
        X[i], eps[i]+1e-15, p=np.inf))-1 for i in range(N)])
    n_y = np.array([len(tree_y.query_ball_point(
        Y[i], eps[i]+1e-15, p=np.inf))-1 for i in range(N)])
    mi = digamma(k) - np.mean(digamma(np.maximum(n_x,1))
        + digamma(np.maximum(n_y,1))) + digamma(N)
    return max(0, mi / np.log(2))

# Generate 3000 sightlines, degrade to resolution R,
# compute per-pixel MI averaged over 80 wavelength bins,
# multiply by n_eff = Delta_lambda / (lambda/R)
```

Listing 1: Toy Model 1: Simulation-Based Mutual Information vs. Resolution

```
import numpy as np

def channel_capacity_vs_redshift():
    """Compute channel capacity as function of source redshift."""
    z = np.linspace(0.1, 7, 200)
    # Effective optical depth (Faucher-Giguere et al. 2008 fit)
    tau_eff = 0.0015 * (1 + z)**4.3
    transmission = np.exp(-tau_eff)
    transmission = np.clip(transmission, 0.001, 1)
    SNR_base = 50
    # Shannon capacity per resolution element
    C = np.log2(1 + SNR_base * transmission) * transmission
```

```python
    return z, C
```

Listing 2: Toy Model 2: Channel Capacity vs. Redshift

```python
import numpy as np

def fisher_info_xHI(z_obs=6.0, sigma_F=0.01):
    """Compute Fisher information for x_HI at given redshift."""
    x_HI = np.logspace(-5, -0.3, 200)
    A_GP = 4.6e5  # Gunn-Peterson coefficient
    # Optical depth
    tau = A_GP * x_HI * (1 + z_obs)**1.5 / 1e5
    F_flux = np.exp(-np.clip(tau, 0, 15))
    # Derivative dF/dx_HI
    dF_dxHI = -A_GP * (1 + z_obs)**1.5 / 1e5 * F_flux
    # Fisher information
    Fisher = dF_dxHI**2 / sigma_F**2
    Fisher = Fisher / Fisher.max()  # normalize
    return x_HI, Fisher
```

Listing 3: Toy Model 3: Fisher Information for Neutral Fraction